

vSECC OCPP 1.6 Compliance Test Report

Vector vSECC.single Board vs MEA CSMS (rddQC4000001)

MEA-V2G Project — Automated Test

2026-06-05 15:51

Contents

1 Test Configuration

Device Under Test	Vector vSECC.single Board
Device IP	192.168.1.166
OCPP Version	1.6 (JSON)
CP ID	rddQC4000001
CSMS	wss://ocpp.measandbox.com:2930
Test PC	192.168.1.100 (WS proxy)
Test Date	2026-06-05 15:51

Test Setup

The vSECC connects to a local WebSocket proxy on the test PC (`ws://192.168.1.100:9000`). The proxy transparently forwards all OCPP frames to the MEA CSMS over TLS (`wss://ocpp.measandbox.com:2930/EV/Srv/JSON/1.6/rddQC4000001`). OCPP extension messages unsupported by MEA (e.g. `SecurityEventNotification`) are intercepted by the proxy and answered with an empty ACK to maintain the session.

2 Test Results Summary

#	Test Item	Detail	Result
Summary: 16/19 PASS 3 FAIL 6 SKIP			
0. vSECC Authentication			
1	vSECC login		PASS
1. Configure vSECC OCPP → MEA CSMS			
2	CSMS URL (local proxy) already = 'ws://192.168.1.100:9000/EV/Srv/JSON/1.6'		SKIP
3	URL identity overwrite (CP ID) already = 'rddQC4000001'		SKIP
4	Identity (CP ID) already = 'rddQC4000001'		SKIP
5	Clear basic auth password already = ''		SKIP
6	Security profile (0=no TLS for ws://) already = '0'		SKIP
7	Backend communication activated already = 'true'		SKIP
8	CSMS URL set correctly	<code>ws://192.168.1.100:9000/EV/Srv/JSON/1.6/rddQC4000001</code>	PASS
9	CP identity set correctly	rddQC4000001	PASS
2. Restart vSECC OCPP Stack			
10	Restart accepted		PASS
3. Wait for OCPP Connection to MEA CSMS			
11	vSECC connected to MEA CSMS		PASS
4. Boot & Heartbeat Verification (via CSMS GetConfig)			
12	GetConfiguration(HeartbeatInterval) → 200	HTTP 200	PASS
13	GetConfiguration(V2GMode) → 200	HTTP 200	PASS
5. Configuration Control			
14	ChangeConfiguration(V2GMode=true) → 200	HTTP 200	PASS

#	Test Item	Detail	Result
15	ChangeConfiguration(MEA_V2G_PowerDemand → 200)	HTTP 200	PASS
16	ChangeConfiguration(MEA_V2G_PowerDemand → 3000/V2G discharge) → 200	HTTP 200	PASS
6. Remote Start/Stop			
17	RemoteStartTransaction → 200	HTTP 200	PASS
18	RemoteStopTransaction → 200	HTTP 200	PASS
7. Reservation			
19	ReserveNow → 200	HTTP 200	PASS
20	CancelReservation → 200	HTTP 200	PASS
8. Smart Charging			
21	SetChargingProfile → 200	HTTP 200	PASS
9. Trigger Messages			
22	TriggerMessage(Heartbeat) → 200	None	FAIL
23	TriggerMessage(MeterValues) → 200	None	FAIL
24	TriggerMessage(StatusNotification) → 200	None	FAIL
10. Reset			
25	Reset(Soft) → 200	HTTP 200	PASS

3 OCPP Message Trace (Proxy Log)

```

1  client = mqtt.Client()
2  mc = mqtt.Client()
3  =====
4  vSECC OCPP Compliance Test vs MEA CSMS
5  CP ID : rddQC4000001
6  CSMS  : ws://192.168.1.100:9000/EV/Srv/JSON/1.6/rddQC4000001
7  =====
8
9  [MQTT monitor started]
10
11
12  0. vSECC Authentication
13
14  [MQTT] vsecc/ocpp_active_network_profile_slot = 1
15  [MQTT] vsecc/ocpp_connection_status = connected
16  [MQTT] vsecc/connector/0/status/availability = operative
17  [MQTT] vsecc/connector/0/status/reserved = false
18  [MQTT] vsecc/connector/0/status/components/active_failures = []
19  [MQTT] vsecc/connector/1/status/charging_authorization_state = pending
20  [MQTT] vsecc/connector/1/status/availability = operative
21  [MQTT] vsecc/connector/1/status/reserved = false
22  [MQTT] vsecc/connector/1/status/components/active_failures = []
23  [PASS] vSECC login
24
25
26  1. Configure vSECC OCPP MEA CSMS
27
28  [SKIP] CSMS URL (local proxy) already = 'ws://192.168.1.100:9000/EV/Srv/JSON/1.6'
29  [SKIP] URL identity overwrite (CP ID) already = 'rddQC4000001'
30  [SKIP] Identity (CP ID) already = 'rddQC4000001'
31  [SKIP] Clear basic auth password already = ''

```

```
32 [SKIP] Security profile (0=no TLS for ws://) already = '0'
33 [SKIP] Backend communication activated already = 'true'
34
35 Verification:
36 [PASS] CSMS URL set correctly (ws://192.168.1.100:9000/EV/Srv/JSON/1.6)
37 [PASS] CP identity set correctly (rddQC4000001)
38
39
40 2. Restart vSECC OCPP Stack
41
42 Sending restart request...
43 [PASS] Restart accepted
44 Waiting 15s for vSECC to reboot...
45 [MQTT] vsecc/ocpp_active_network_profile_slot = 1
46 [MQTT] vsecc/ocpp_connection_status = connecting
47 [MQTT] vsecc/connector/1/status/reserved = false
48 [MQTT] vsecc/ocpp_connection_status = connected
49 [MQTT] vsecc/connector/0/status/availability = inoperative
50 [MQTT] vsecc/connector/0/status/reserved = false
51 [MQTT] vsecc/connector/1/status/charging_authorization_state = pending
52 [MQTT] vsecc/connector/1/status/availability = inoperative
53 [MQTT] vsecc/connector/1/status/reserved = false
54 [MQTT] vsecc/connector/1/status/components/active_failures = []
55 [MQTT] vsecc/connector/0/status/components/active_failures = []
56 [MQTT] vsecc/connector/0/status/availability = operative
57 [MQTT] vsecc/connector/0/status/reserved = false
58 [MQTT] vsecc/connector/1/status/charging_authorization_state = pending
59 [MQTT] vsecc/connector/1/status/availability = operative
60 [MQTT] vsecc/connector/1/status/reserved = false
61 [MQTT] vsecc/connector/1/status/charging_standard =
62
63
64 3. Wait for OCPP Connection to MEA CSMS
65
66 Waiting up to 90s for vSECC to connect...
67 vSECC OCPP status: connected (via MQTT)
68 [PASS] vSECC connected to MEA CSMS
69
70
71 4. Boot & Heartbeat Verification (via CSMS GetConfig)
72
73 [PASS] GetConfiguration(HeartbeatInterval) 200 (HTTP 200)
74 [PASS] GetConfiguration(V2GMode) 200 (HTTP 200)
75     V2GMode = not present
76
77
78 5. Configuration Control
79
80 [PASS] ChangeConfiguration(V2GMode=true) 200 (HTTP 200)
81 [PASS] ChangeConfiguration(MEA_V2G_PowerDemand=3000) 200 (HTTP 200)
82 [PASS] ChangeConfiguration(MEA_V2G_PowerDemand=-3000/V2G discharge) 200 (HTTP 200)
83
84
85 6. Remote Start/Stop
86
87 [PASS] RemoteStartTransaction 200 (HTTP 200)
88 [PASS] RemoteStopTransaction 200 (HTTP 200)
89
90
91 7. Reservation
92
93 [PASS] ReserveNow 200 (HTTP 200)
94 [PASS] CancelReservation 200 (HTTP 200)
```

8. Smart Charging

[PASS] SetChargingProfile 200 (HTTP 200)

9. Trigger Messages

[FAIL] TriggerMessage(Heartbeat) 200 (None)

[FAIL] TriggerMessage(MeterValues) 200 (None)

[FAIL] TriggerMessage(StatusNotification) 200 (None)

10. Reset

[PASS] Reset(Soft) 200 (HTTP 200)

Test Complete

MQTT events captured: 109

Recent MQTT events:

[MQTT] vsecc/connector/1/ev/time_to_full_soc =
[MQTT] vsecc/connector/1/ev/time_to_bulk_soc =
[MQTT] vsecc/connector/1/ev/v2g_session_id =
[MQTT] vsecc/connector/1/ev/minimum_soc =
[MQTT] vsecc/connector/1/ev/target_soc =
[MQTT] vsecc/connector/1/ev/bulk_soc =
[MQTT] vsecc/connector/1/ev/full_soc =
[MQTT] vsecc/connector/1/ev/charge_mode_info =
[MQTT] vsecc/connector/1/ev/supported_schemas =
[MQTT] vsecc/connector/1/ev/pp_state = disconnected
[MQTT] vsecc/connector/1/ev/cp_max_voltage = 11.600905
[MQTT] vsecc/connector/1/ev/cp_state = state_a
[MQTT] vsecc/config_change = df68a2d9
[MQTT] vsecc/io/a_in/1/value = 0.02
[MQTT] vsecc/io/d_out/14/value = 0